

Software Architecture Document

Folder Structure

expense_tracker/

├── main.py

├── expense.py

├── storage.py

├── reports.py

├── utils.py

├── expenses.csv

└── README.md

Module Responsibilities

main.py

- Program entry point
 - Menu display
 - User interaction
 - Navigation
-

expense.py

Represents a single expense.

Attributes:

- date
- category
- description
- amount

Responsible for validating expense data.

storage.py

Handles:

- Loading CSV
- Saving CSV
- Adding records
- Deleting records

This module is the only one allowed to interact with files.

reports.py

Responsible for:

- Monthly summaries
 - Category totals
 - Searching expenses
-

utils.py

Contains helper functions:

- Input validation
 - Currency formatting
 - Date parsing
 - Menu formatting
-

Data Flow

User Input

↓

main.py

↓

Business Logic

↓

storage.py

↓

expenses.csv

Design Principles

- Single Responsibility Principle
 - Separation of Concerns
 - Modular Design
 - Low Coupling
 - High Cohesion
-

Error Handling Strategy

Every user input shall be validated.

Invalid numeric input shall prompt the user again.

Missing files shall be created automatically.

Unexpected exceptions shall display an error message without terminating the application.

Coding Standards

- snake_case naming
- descriptive function names
- functions under ~30 lines where practical
- no duplicated code
- use docstrings for public functions
- organize imports consistently